Swap Regret under Best-Response Play

In our setting, players interact with a moderator. At time t, the moderator samples a joint recommendation $R^{(t)} \sim \mathbf{x}^{(t)} \in \Delta(A)$ and sends each player i the private recommendation $R_i^{(t)}$ along with $\mathbf{x}^{(t)}$. Player i analyses the information $(\mathbf{x}^{(t)}, R_i^{(t)})$ and chooses to play $A_i^{(t)}$.

Regret of a player. The swap regret for player i is:

$$\mathcal{R}_i := \max_{\psi: A_i \to A_i} \sum_{t=1}^{T} \mathbb{E}\big[u_i(\psi(A_i^{(t)}), A_{-i}^{(t)}) - u_i(A_i^{(t)}, A_{-i}^{(t)})\big], \tag{1}$$

where the expectation is over the sampling $R^{(t)} \sim \mathbf{x}^{(t)}$.

First scenario: known utilities

The moderator samples $R^{(t)} \sim \mathbf{x}^{(t)} \in \mathrm{CE}(G)$. Since $\mathbf{x}^{(t)}$ is a CE, under BR no player deviates, so $A^{(t)} = R^{(t)}$. Then for any ψ:

$$\mathbb{E}\big[u_i(\psi(A_i^{(t)}), A_{-i}^{(t)}) - u_i(A_i^{(t)}, A_{-i}^{(t)})\big] = \mathbb{E}_{R_i}\Big[\mathbb{E}_{R_{-i}\sim\mathbf{x}(\cdot|R_i)}\big[u_i(\psi(R_i^{(t)}), R_{-i}^{(t)}) - u_i(R_i^{(t)}, R_{-i}^{(t)})\big]\Big].$$

Since $\mathbf{x}$ is a CE, $\mathbb{E}_{R_{-i}\sim\mathbf{x}(\cdot|R_i)}[u_i(\psi(R_i), R_{-i}) - u_i(R_i, R_{-i})] \le 0$ for all ψ and all R_i. Therefore $\mathcal{R}_i = 0$.

Second scenario: unknown game

The moderator runs Algorithm 5 of [1], producing $\mathbf{x}^{(t)} \in \mathrm{CE}(\hat{G}^{(t)})$ (a CE of the estimated game). Utilities are normalized to $[0, 1]$.

For a fixed swap function $\psi : A_i \to A_i$, define:

$$\Delta_i(A^{(t)}) := u_i(\psi(A_i^{(t)}), A_{-i}^{(t)}) - u_i(A_i^{(t)}, A_{-i}^{(t)}), \qquad \Delta_i(R^{(t)}) := u_i(\psi(R_i^{(t)}), R_{-i}^{(t)}) - u_i(R_i^{(t)}, R_{-i}^{(t)}).$$

When $A^{(t)} = R^{(t)}$: $\Delta_i(A^{(t)}) = \Delta_i(R^{(t)})$. Otherwise, since $u_i \in [0, 1]$: $|\Delta_i(A^{(t)}) - \Delta_i(R^{(t)})| \le 2 \cdot \mathbf{1}\{A^{(t)} \neq R^{(t)}\}$.

So:

$$\mathbb{E}[\Delta_i(A^{(t)})] \le \underbrace{\mathbb{E}[\Delta_i(R^{(t)})]}_{\text{(I): recommendation swap regret}} + \underbrace{2\,\Pr(A^{(t)} \neq R^{(t)})}_{\text{(II): deviation penalty}}. \tag{2}$$

Bounding (I).

$$\mathbb{E}[\Delta_i(R^{(t)})] = \sum_{r_i} \sum_{r_{-i}} \mathbf{x}^{(t)}(r_i, r_{-i})\big[u_i(\psi(r_i), r_{-i}) - u_i(r_i, r_{-i})\big] = \sum_{r_i} \varphi_i(r_i, \psi(r_i), \mathbf{x}^{(t)}),$$

where $\varphi_i(a_i, a_i', \mathbf{x}) := \sum_{a_{-i}} \mathbf{x}(a_i, a_{-i})[u_i(a_i', a_{-i}) - u_i(a_i, a_{-i})]$. Since $\mathbf{x}^{(t)} \in \mathrm{CE}(\hat{G}^{(t)})$:

$$\varphi_i(a_i, a_i', \mathbf{x}^{(t)}) \le \overset{\mathbf{x}^{(t)}}{\Pr}(a_i) \cdot \varepsilon_t,$$

where $\varepsilon_t := \max_{i, a_i, a_i'} \|\mathbf{w}_i^* - \mathbf{w}_i^{(t)}\|_\infty$ is the estimation error. Therefore:

$$\sum_{r_i} \varphi_i(r_i, \psi(r_i), \mathbf{x}^{(t)}) \le \sum_{r_i} \overset{\mathbf{x}^{(t)}}{\Pr}(r_i)\, \varepsilon_t = \varepsilon_t.$$

Summing over t: $\displaystyle\sum_{t=1}^{T} \mathbb{E}[\Delta_i(R^{(t)})] \le \sum_{t=1}^{T} \varepsilon_t = O(nM \log T)$.

Bounding (II). Under γ-BR (players deviate only when the gain exceeds $\gamma > 0$), if $A_j^{(t)} \neq R_j^{(t)}$ then $\varphi_j(R_j^{(t)}, A_j^{*(t)}, \mathbf{x}^{(t)}) > \gamma$, so:

$$\mathbf{1}\{A_j^{(t)} \neq R_j^{(t)}\} \leq \frac{\varphi_j(R_j^{(t)}, A_j^{*(t)}, \mathbf{x}^{(t)})}{\gamma}.$$

By a union bound $\mathbf{1}\{A^{(t)} \neq R^{(t)}\} \leq \sum_j \mathbf{1}\{A_j^{(t)} \neq R_j^{(t)}\}$, so:

$$\sum_{t=1}^{T} \Pr(A^{(t)} \neq R^{(t)}) \leq \frac{1}{\gamma} \sum_{t=1}^{T} \sum_j \mathbb{E}\big[\varphi_j(R_j^{(t)}, A_j^{*(t)}, \mathbf{x}^{(t)})\big] = \frac{\mathrm{Reg}(\mathcal{A})}{\gamma},$$

where $\mathrm{Reg}(\mathcal{A}) = O(nM \log T)$ is the compliance regret (Theorem 8 of [1]).

Conclusion. Combining (I) and (II) in (7Decompositionequation.4.7), maximizing over ψ, and summing over t:

$$\boxed{\mathcal{R}_i \leq O(nM \log T) + \frac{2}{\gamma} \mathrm{Reg}(\mathcal{A}) = O\left(\left(1 + \frac{1}{\gamma}\right) nM \log T\right).}$$

Reference. [1] Alanqary, Baba, Wu, Bayen. *Learning to Recommend in Unknown Games.* arXiv:2602.16998, 2026.